The Pop Bottle Magnetometer

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The Concept:

A pop bottle magnetometer shows the direction of magnetic north using a laser pointer. The laser beam reflects onto a nearby wall, creating a small spot of light. When a geomagnetic storm occurs, the location of magnetic north temporarily shifts. This causes the laser spot to move on the wall. By recording the movement of the laser spot over time, we can monitor geomagnetic activity.

This project works well as a longer-term investigation for students. The magnetometer itself can be constructed within a single one-hour class, but you will need to allow several weeks for students to record data. An additional class session should then be set aside for plotting and analysing the results.

Aim:

- Construct a simple magnetometer using a plastic pop bottle and understand how it can be used to monitor changes in Earth's magnetic field.
- Learn how to retrieve and plot data from the magnetometer in order to identify geomagnetic storm events.

Australian Curriculum: Science F-10 Version 9.0 links:

Select and use equipment to generate and record data with precision, using digital tools as appropriate (AC9S7I03 AC9S8I03).

Analyse data and information to describe patterns, trends and relationships and identify anomalies (AC9S7I05 AC9S8I05).

Materials (Figure 1):

- Colourless 2 litre plastic bottle
- N-S bar magnet e.g., from aussie magnets
- Piece of mirror card
- Cotton sewing thread (dental floss/ fishing line for an improved design)
- Piece of cardboard

- Straw
- Scissors
- Sellotape
- Uncooked rice/ lentils or sand
- Field compass e.g., from anaconda
- Laser pointer e.g., from ebay
- Desk lamp (optional)

Figure 1: Materials for the pop bottle magnetometer project.

Safety considerations:

- Keep strong magnets away from credit cards, mobile phones, computers and other technology to avoid potential damage.
- Handle scissors and nails/ sewing needles carefully to avoid injury.
- The raw cut edge of the plastic bottle may be sharp. Take care when handling this part of the bottle.

How to make your magnetometer:

Demonstration Video: The Pop Bottle Magnetometer: How to Make & Set Up a Pop Bottle Magnetometer

1. Clean your plastic bottle and remove any labels.
2. Use scissors to carefully cut off the top 1/3 of the plastic bottle. This may be easier with the cap of the bottle removed.
3. Half fill the bottom part of the plastic bottle with sand or uncooked rice/ lentils to stabilise it (see **Figure 2**).
4. Cut a straw to a length just slightly shorter than the length of the bar magnet. Secure the straw to the long edge of the bar magnet using pieces of sellotape.
5. Cut a rectangular piece of cardboard around 10 cm long and with a width the same as the length of the bar magnet.
6. Stick your magnet to the top of the piece of card using sellotape, as shown in **Figure 3**.
7. Stick a piece of mirror card (with a width equal to the width of the magnet) to the centre of the magnet, as shown in **Figure 3**.
8. Cut around 30 cm of cotton sewing thread. Thread the end through the straw and tie it back to itself as a triangular loop with equal sides a few cm in length, as shown in **Figure 3**.
9. Check that the magnet hangs vertically. Adjust the length of the sides of the triangular loop of thread if needed.
10. Remove the cap of the bottle and <u>carefully</u> pierce a small hole through it using a nail/ sewing needle.
11. Thread the other end of the cotton through the top of the bottle, then through the hole in the bottle top. Place the cap back onto the bottle.
12. Reattach the top part of the bottle to the bottom using pieces of sellotape. The card with the magnet should now be inside the bottle.
13. Hang the magnet inside of the bottle by adjusting the length of the thread. Then tape the thread to the side of the bottle cap to secure it.

Your finished magnetometer should look like **Figure 4.**

Figure 2: Cleaned, clear pop bottle with the top 1/3 cut off and the bottom part half-filled with rice.

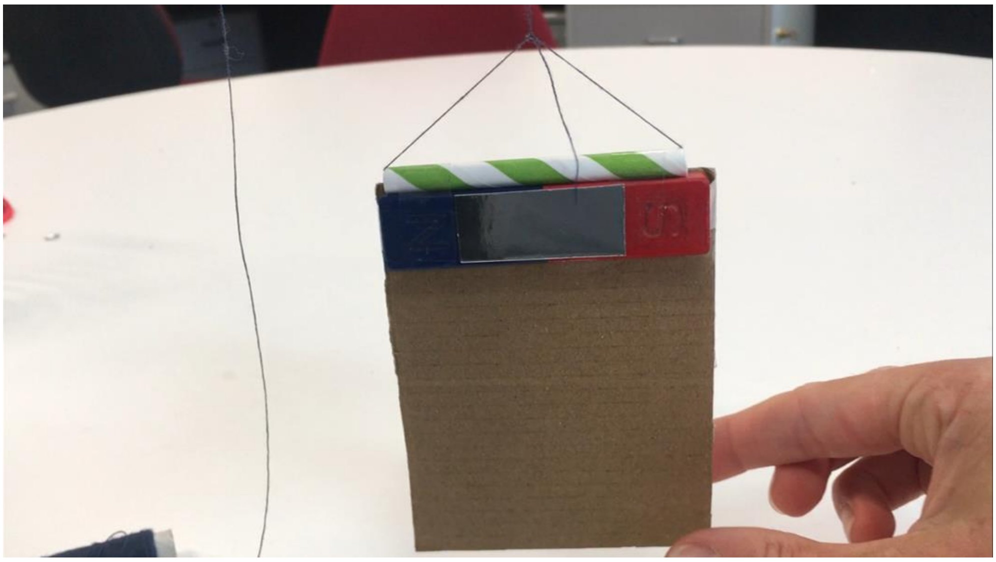

Figure 3: Bar magnet with the straw stuck to the top and mirror card stuck to the centre. These are taped to a piece of cardboard ~10cm long. Cotton thread is threaded through the straw and tied as a triangular loop with sides of equal length (a few cm).

Figure 4: Completed pop bottle magnetometer.

Check that:

- The magnet is hanging vertically and can swing freely.
- The sides of the cardboard piece do not touch the sides of the bottle when it swings.
- The bottom of the carboard piece does not touch the sand/ rice/ lentils at the bottom of the bottle. Trim the length of the cardboard piece or remove some of the sand/ rice if needed.
- The card is suspended securely, and the thread holding it does not slip where it is secured to the bottle cap.

Checking that your magnetometer is working:

Demonstration video: The Pop Bottle Magnetometer: How to Make & Set Up a Pop Bottle Magnetometer

You can check that your magnetometer is working by allowing the magnet to stabilise and then checking its orientation with a field compass. The long edge of the magnet/ short edge of the piece of card should be aligned with the compass needle (see **Figure 5**).

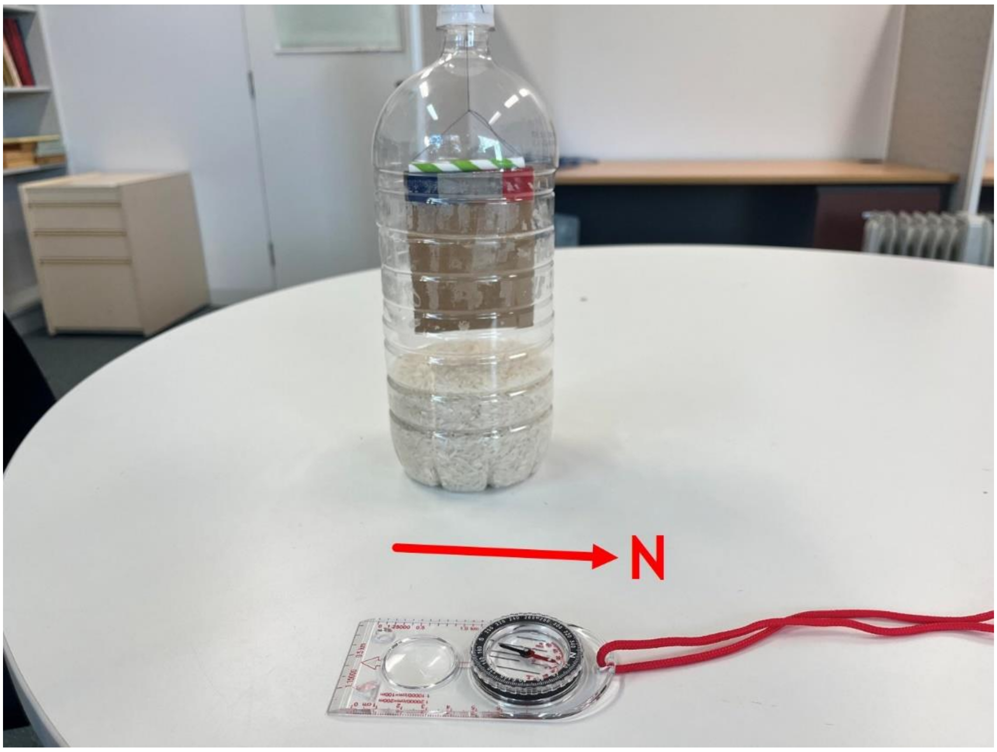

Figure 5: Pop bottle magnetometer and field compass both point north.

Setting up your magnetometer:

Demonstration video: [The Pop Bottle Magnetometer: How to Make & Set Up a Pop Bottle Magnetometer]

- Locate a quiet, undisturbed area to set up your pop bottle magnetometer, away from electronic/ moving metal objects. Ideally, it should be placed in a room that no-one will enter during the recording period.
- Place the magnetometer on a table facing a wall around 1 metre away (see **Figure 6**). Make a note of the exact distance.
- Point the laser pointer at the magnetometer and adjust its height so that the laser spot hits the centre of mirror card on the magnet (see **Figure 7**). You may need to place it on top of a sturdy box/ pile of books to get the correct height. You may also need to leave your laser plugged in to the mains electricity depending on its battery life.
- The reflected laser spot should be seen on the wall as a small, clear spot.
- The distance between the laser and the bottle is not important, but there should be around a 20° angle between the laser and the mirror on the magnet to get a good reflection (AuroraWatch, University of Alberta, 2007). Adjust this angle/ distance to adjust the laser spot size/ shape.

- Stick a ruler onto the wall where the spot of light is. Make sure the spot is above the centre of the ruler – this is the reference position (see **Figure 8**).
- If you are filming a time lapse of your magnetometer, place a phone on a tripod in front of the wall at a height where the ruler and laser spot are clearly in view.

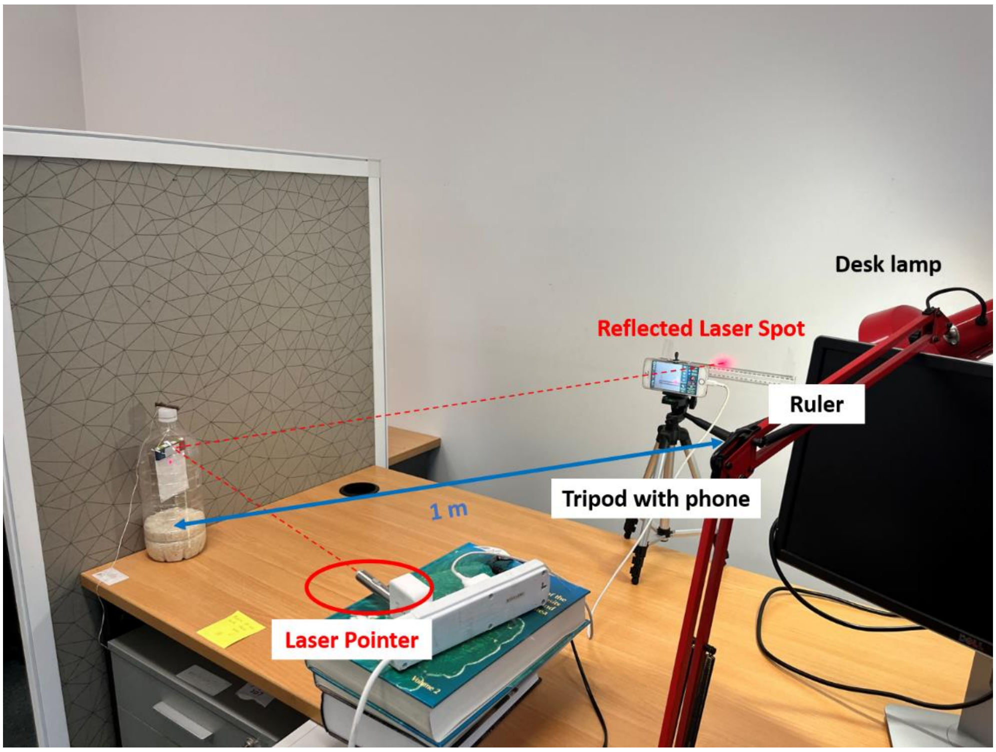

Figure 6: Pop bottle magnetometer set up. The centre of the bottle is 1 metre away from the reflected spot on the wall, and the laser pointer is pointed at the centre of the mirror on the magnet.

Figure 7: Laser pointer hits the centre of the mirror card on the magnet.

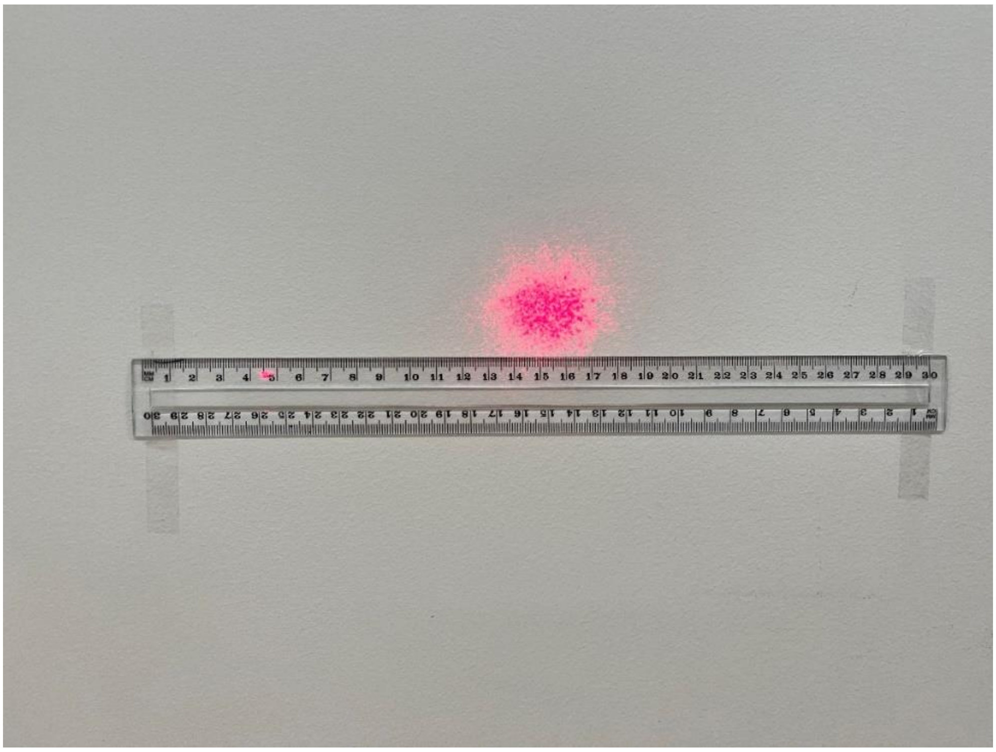

Figure 8: Reflected spot of light in the centre of the ruler stuck to the wall. In this example, 15 cm (the centre of the ruler) is the reference position.

Improvements to the design & set up:

Demonstration video: The Pop Bottle Magnetometer: How to Make & Set Up a Pop Bottle Magnetometer

The following improvements can be made to the magnetometer design/ set up:

- If the position of the reflected spot is drifting down the wall over time, this could be due to the sewing thread untwisting/ stretching. Try using a non-stretch thread like dental floss or fishing line (see **Figure 9**).
- Drifting could also be due to movement of the laser. Ensure it is securely fixed in place to a solid surface.
- If the thread is slipping, try securing it to the outside of the bottle top by tying it around a small dowel/ stick (see **Figure 9**).
- If the magnet isn't hanging horizontally, you can stick small pieces of blue tac to the bottom corners of the card to balance it (see **Figure 10**).
- You may need to illuminate the wall with a desk lamp if you set up your magnetometer in a room where the light levels are changeable. The laser spot and ruler need to be visible every hour.

Figure 9: The magnet can be suspended by dental floss to reduce stretch. Securing the floss by tying it around a small stick on top of the bottle cap reduces slip.

Figure 10: Blu tac in the corners of the card to help balance it and make sure it hangs vertically.

Time lapsing the laser spot:

The best way to record the movement of the reflected laser spot on the wall is by filming a timelapse video. Taking a photo once an hour during the recording period is sufficient to show changes in Earth's magnetic field.

Example pop bottle magnetometer timelapses can be watched here: The Pop Bottle Magnetometer: Example Timelapses & Data

The Lapse It app on an iPhone SE was used to time the pop bottle magnetometer in this project. This app is available from the Apple App Store and Google Play Store.

- Place a mobile phone with the Lapse It app open on a tripod in front of the ruler on the wall.
- Adjust the height of the tripod to make sure that the laser spot and ruler are clearly visible in the frame.
- Tap on the screen to set the focus.
- Set the interval to 1 Hour.
- Make sure ‘Auto Stop’ is disabled.
- Press ‘Prepare capture’ to begin recording.
- At the end of the recording period press ‘Finish capture’.

- Check the option to display timestamps on the frames. This shows the date/ time of each image in DD/MM/YYYY HH:MM:SS format in the bottom left corner of the video.
- You can also name your timelapse and display the title on the video.
- You can change the run time of your timelapse video using ‘Sequence Duration’.
- To download your individual timelapse images, select ‘Frames Catalog’ then ‘Export as Zip’. This downloads a zip file containing all your images to your device (a paid subscription is required, $7.99 AUD a year or $2.99 AUD for 3 months).
- Image file names are in the format YYYYMMDD_HHMMSS[frame number].
- If you do not wish to pay for the app, you can take screenshots of each frame from the video timelapse.

Processing the data from your timelapse:

- To get data from your timelapse, record the position of the centre of the laser spot in each timelapse frame.
- You can import the images into a PowerPoint file and use a shape over the centre of the laser spot to make this easier. See the example below (**Figure 11**):

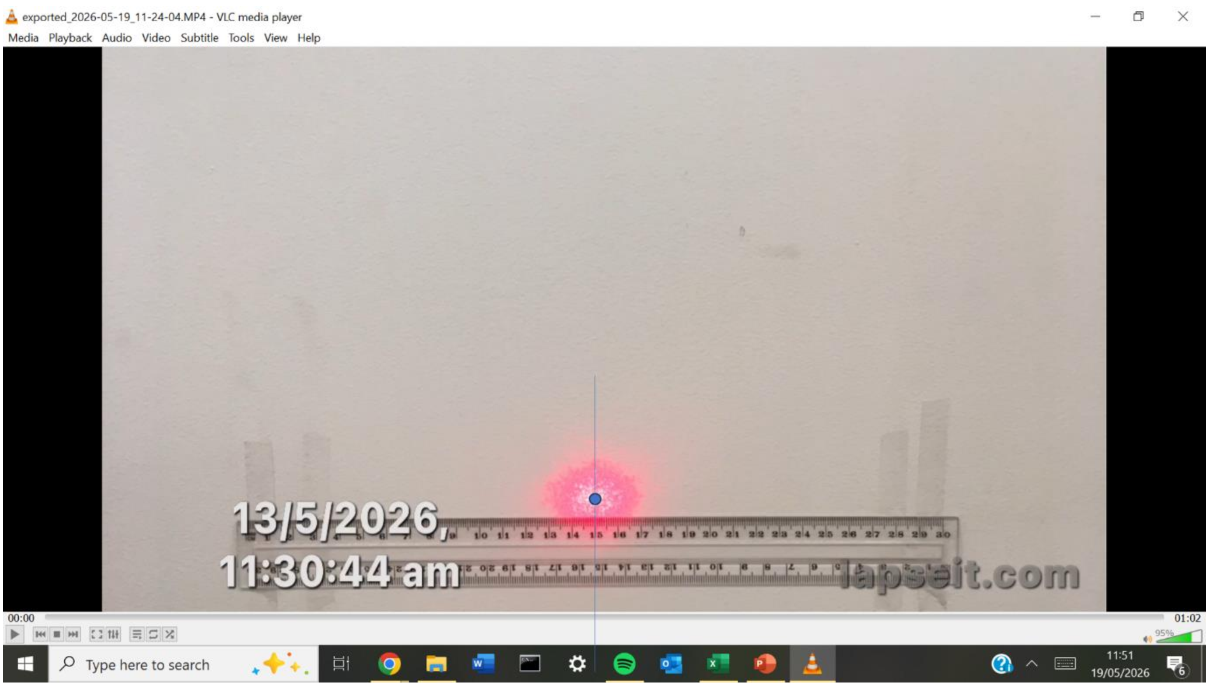

Figure 11: A frame from a pop bottle magnetometer timelapse with a blue shape added to the centre of the laser spot in Microsoft PowerPoint to make tracking changes in the spot position easier.

- Record the spot position over time in an Excel spreadsheet. Try to record to at least the nearest 5 mm.
- Plot the spot position over time as a time series graph. This graph is a known as a **magnetogram**.

Identifying geomagnetic storm events in your magnetogram:

Geomagnetic storm events will be seen as spikes in your magnetogram. Spikes show where the spot has moved far from the reference position in response to a shift in the position of magnetic north. If geomagnetic activity levels are quiet (no storms), the laser spot will remain relatively stationery around the reference position.

Note: Space weather events are infrequent. To guarantee recording a geomagnetic storm event, it is a good idea to check the 3-day space weather forecasts from the NOAA here or Australian Bureau of Meteorology here and look for scheduled storm events (G1 +).

You can compare your pop bottle magnetograms to your local geomagnetic observatory by viewing magnetograms from INTERMAGNET here, or downloading data from the Bureau of Meteorology here (Note that data from the Hobart geomagnetic observatory is not available to view on INTERMAGNET).

The horizontal (North facing) component of the magnetic field is the most relevant as a pop bottle magnetometer only records shifts in the location of magnetic North. For the INTERMAGNET data portal, the **X** component faces North. For the BoM Hobart Magnetometer, the **h** component faces North. Make sure to check with the specific observatory you are comparing your data to.

You can also look at the Kp index data from GFZ Helmholtz Centre for Geosciences here to see if a geomagnetic storm event occurred during your recording period. A Kp index of 5 or higher indicates a geomagnetic storm (NOAA). See the NOAA space weather scales for more information.

Examples of stormy and quiet pop bottle magnetometer plots and their comparison to geomagnetic observatory data can be seen in the document

'*The Pop Bottle Magnetometer - Example Data*'.

Calculating degrees of deflection:

Space weather events temporarily distort earth's magnetic field and produce angular shifts in the location of magnetic north (NASA, 2015). This can be detected by your magnetometer, and the displacement of the reflected spot can be converted into degrees of deflection:

For a 1 metre distance:

Degrees of deflection = displacement of reflected point (cm) x 0.25

(NASA, 2015)

Increasing the distance between the centre of the bottle and the reflected spot on the wall allows you to measure smaller solar storm events.

For longer distances (1 – 7 metres):

Degrees of deflection = (1/L) x 0.25

Where L = distance between the centre of the bottle and the reflected spot in metres

(NASA, 2015)

You can calculate degrees of deflection in your Excel spreadsheet containing your spot position over time data and create a second graph of degrees of deflection over time.

According to NASA (2015), strong magnetic storms can cause ≥ 1° of deflection. For more common, weaker storms, degrees of deflection is often 0.5° to 1°.

For a more detailed guide on understanding the data from a pop bottle magnetometer, plus alternative designs, see:

NASA (2015). Soda Bottle Magnetometers for Space Weather Studies. Available at: https://science.nasa.gov/learn/heat/resource/soda-bottle-magnetometers-student-guide/

This activity was inspired by:

AuroraWatch, University of Alberta (2007). *How to use your Pop Bottle Magnetometer*. https://www.aurorawatch.ca/content/view/16/40/

AuroraWatchUK, *Build a Detector*. https://aurorawatch.lancs.ac.uk/detectors/

Met Office (2025). *Make your own magnetometer.* Available at: https://weather.metoffice.gov.uk/learn-about/met-office-for-schools/other-content/magnetometer

NASA (2015). *Soda Bottle Magnetometers for Space Weather Studies.* Available at: https://science.nasa.gov/learn/heat/resource/soda-bottle-magnetometers-student-guide/

NASA (2004). *Earth's Magnetic Personality.* Available at: https://cse.ssl.berkeley.edu/SegwayEd/lessons/exploring_magnetism/pdf_Files.html

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References:

AuroraWatch, University of Alberta. (2007). *How to use your Pop Bottle Magnetometer.* Available at: https://www.aurorawatch.ca/content/view/16/40/

Bureau of Meteorology, Space Weather Services. *Space Weather Conditions.* Available at: https://www.sws.bom.gov.au/

Bureau of Meteorology, Space Weather Services. *World Data Centre for Geomagnetism.* Available at: https://www.sws.bom.gov.au/World_Data_Centre/1/2

INTERMAGNET. *Intermagnet Data Viewer*. Available at: https://imag-data.bgs.ac.uk/GIN_V1/GINForms2

Matzka, J., Stolle, C., Yamazaki, Y., Bronkalla, O. and Morschhauser, A. (2021). The geomagnetic Kp index and derived indices of geomagnetic activity. *Space Weather*. https://doi.org/10.1029/2020SW002641

NASA (2015). *Soda Bottle Magnetometers for Space Weather Studies.* Available at: https://science.nasa.gov/learn/heat/resource/soda-bottle-magnetometers-student-guide/

NOAA Space Weather Prediction Center. *3-Day Geomagnetic Forecast.* Available at: https://www.spaceweather.gov/products/3-day-geomagnetic-forecast

NOAA Space Weather Prediction Center. *NOAA Space Weather Scales.* Available at: https://www.spaceweather.gov/noaa-scales-explanation

This project uses the Lapse It mobile app http://www.lapseit.com/

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